

Predicting Healthcare Insurance costs and Risk Categories Using Machine Learning

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Introduction, Purpose & Aim

Introduction

Health insurance costs vary significantly based on demographics, lifestyle choices, and underlying health risks. Machine learning enables data-driven prediction and classification to improve risk assessment and pricing strategies.

Purpose

Analyze factors influencing insurance charges through regression modeling for cost prediction and classification for risk tier assignment. Compare performance of multiple ML models.

Aim

Identify the most accurate and reliable model while evaluating the impact of feature engineering, data transformation, and advanced ensemble methods for better insurance analytics decision-making.

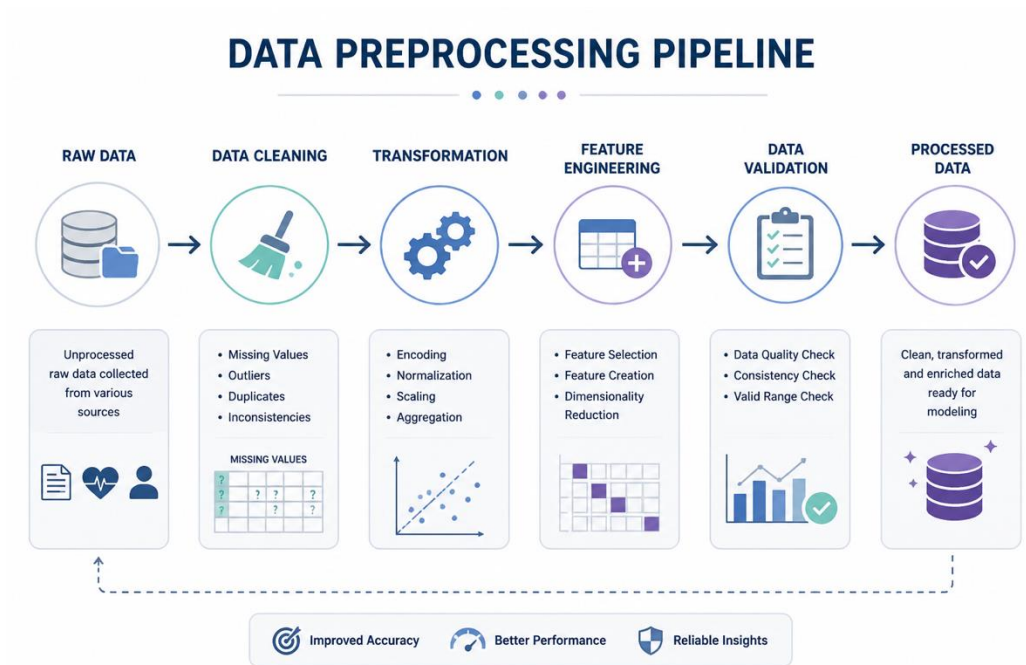
Data Preparation & Initial Inspection

Data Overview

Dataset loaded from CSV with 1,338 initial records cleaned to 1,335 observations. Contains 12 variables covering demographics, health metrics, and insurance charges.

Data Inspection

Verified data types and structure, identified missing values in age, BMI, and children variables. Removed incomplete rows using `dropna()`, resulting in 1,335 clean observations.



Exploratory Data Analysis (EDA)

Main Findings

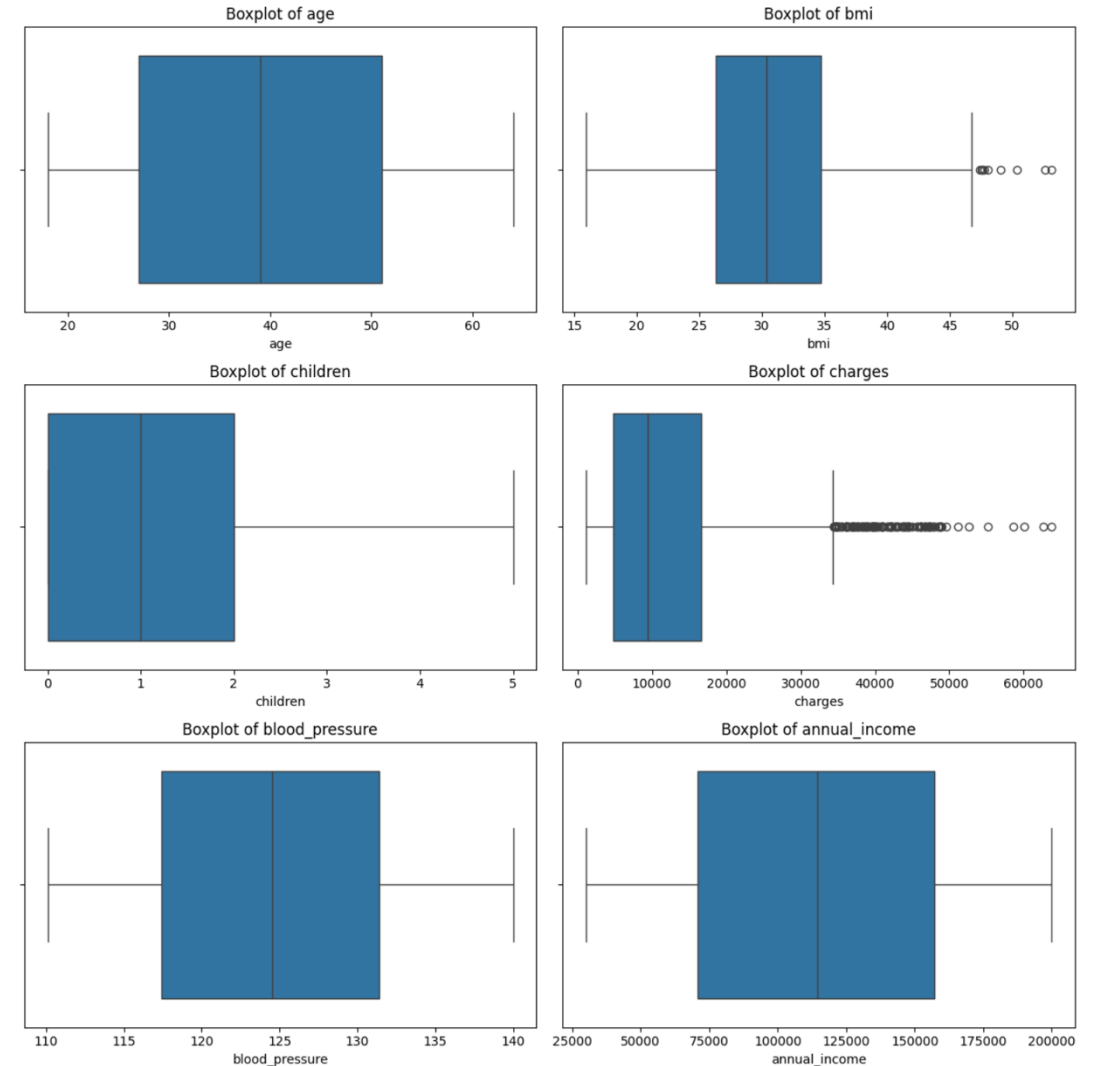
- Average age: 39 years; average BMI: 30.7
- Insurance charges are highly variable, ranging up to about \$63K
- Most individuals are non-smokers
- Most common region: Southeast
- Most common occupation risk: Moderate

Visual Insights

- Histograms show that charges is strongly right-skewed.
- Boxplots confirm outliers in charges and BMI-related variables.
- Other variables such as age, blood pressure, and income are more stable

Key Takeaway

- EDA shows that insurance charges contain skewness and outliers, which supports using log transformation and robust preprocessing before modeling.



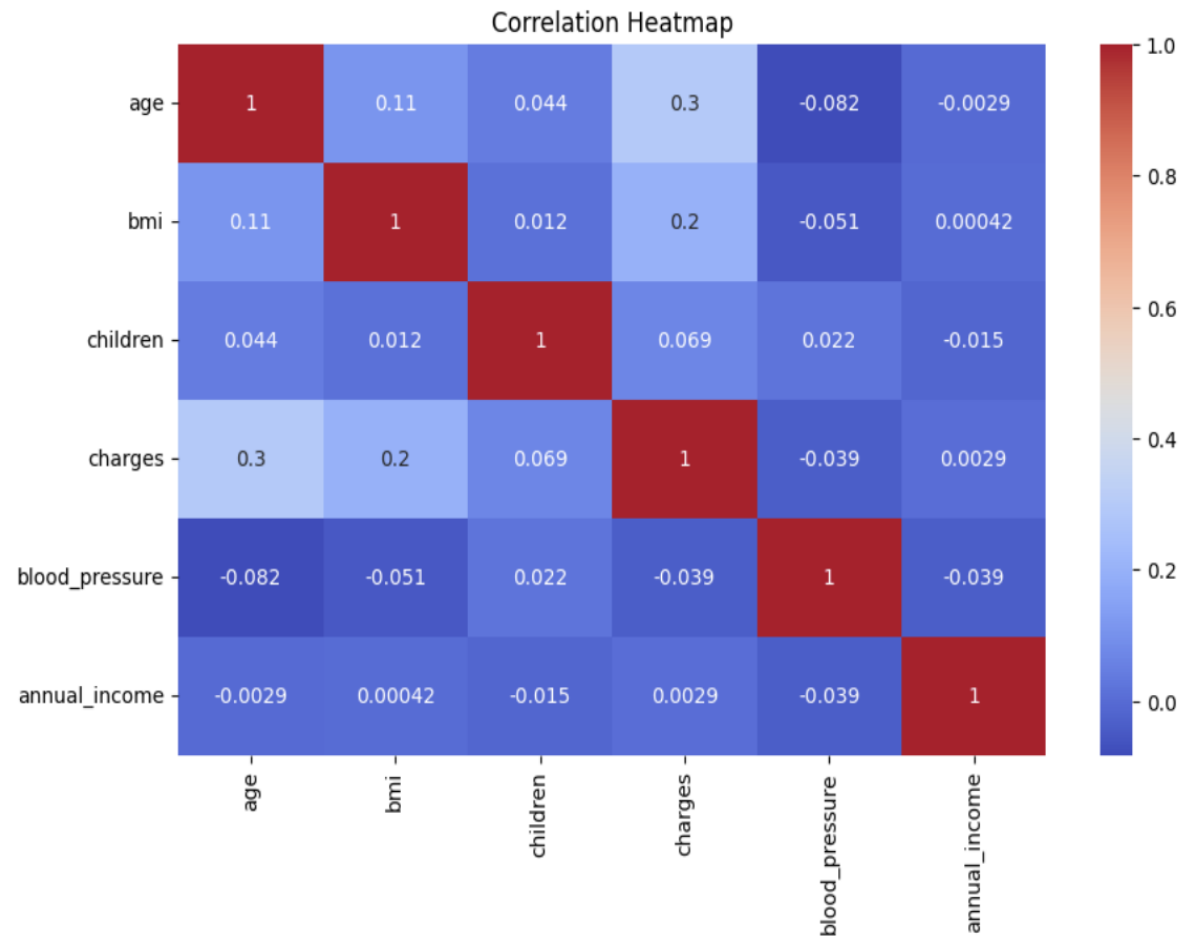
Feature Relationships

Key Findings

- charges vs log_charges: 0.89 → log transformation preserves pattern
- smoker_binary & bmi_smoker_interaction: strong positive relationship with charges
- age vs age_bp_risk: 0.98 → confirms engineered feature design
- annual_income vs charges: ~0.00 → negligible impact
- blood_pressure vs charges: -0.04 → weak relationship

Key Takeaway

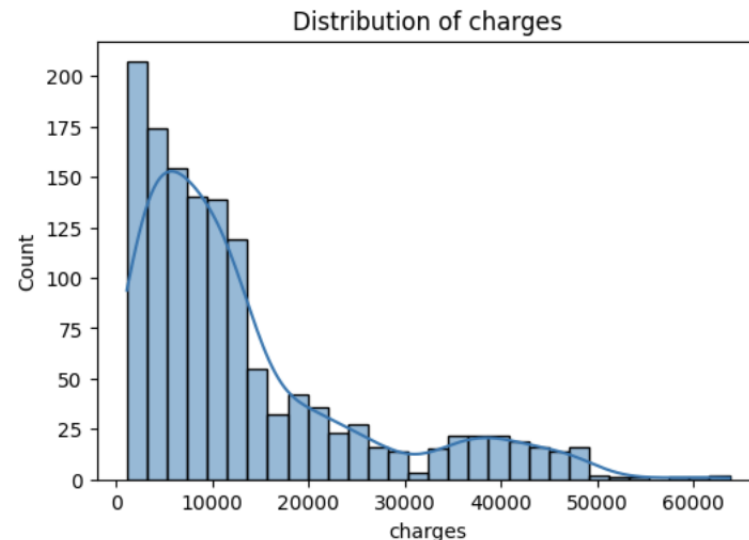
- Insurance costs are primarily driven by smoking behavior and age-related risk, not income



Handling Skewness: Log Transformation

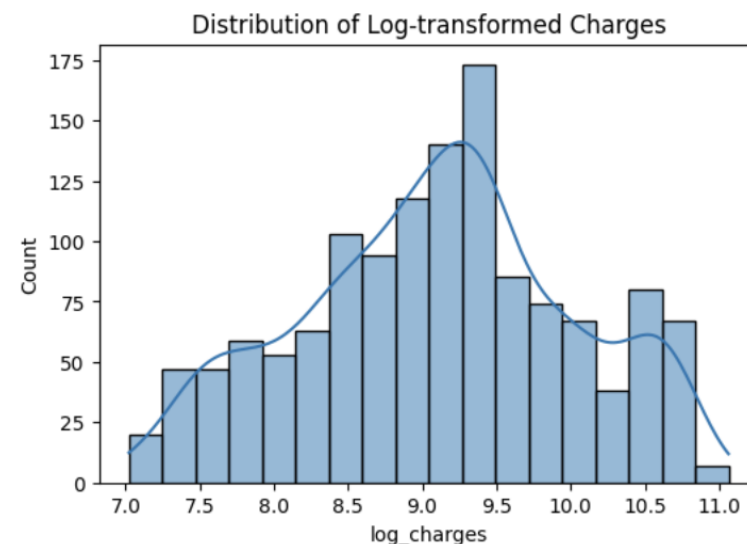
Before Transformation

- Charges range: \$1,121 → \$63,770
- Highly right-skewed distribution
- Mean: 13,255 vs Median: 9,378 → strong skew
- Presence of extreme high-cost outliers



After Transformation

- Log range: 7.0 → 11.0
- Distribution becomes approximately normal
- Reduced impact of extreme values
- Improved stability for regression models



Key Takeaway

- Log transformation reduces skewness and helps models learn better patterns without being dominated by outliers

Feature Engineering & Modeling Setup

Risk Tier Creation

- Quantile-based classification of charges
- Balanced dataset: Low: 445, Medium: 445 , High: 445

Model Setup

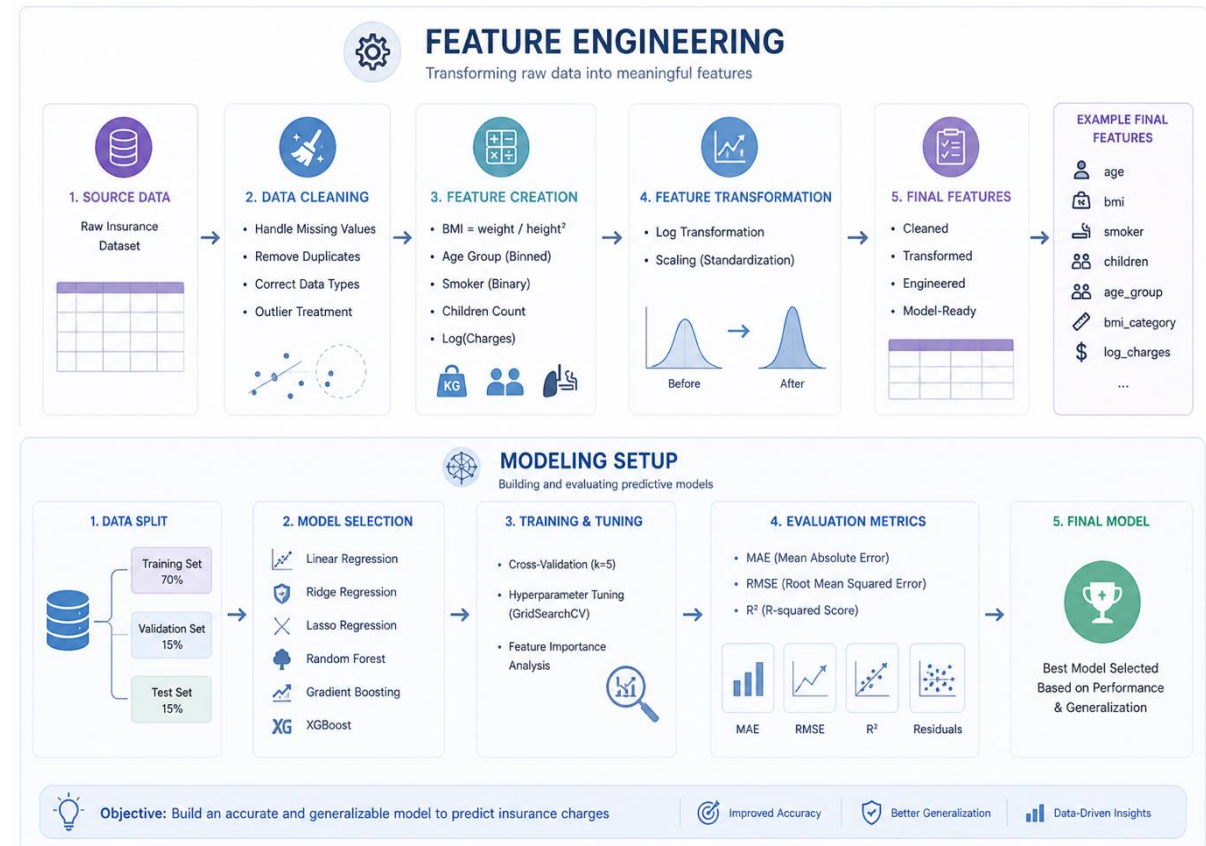
- Regression target: log_charges
- Classification target: risk_tier
- Split: 70% train / 15% validation / 15% test

Preprocessing:

- Numeric: median imputation + scaling
- Categorical: most frequent imputation + one-hot encoding

Key Takeaway

- Balanced, feature-enhanced dataset optimized for modeling.



Principal Component Analysis (PCA) Dimensionality Reduction

Key Findings

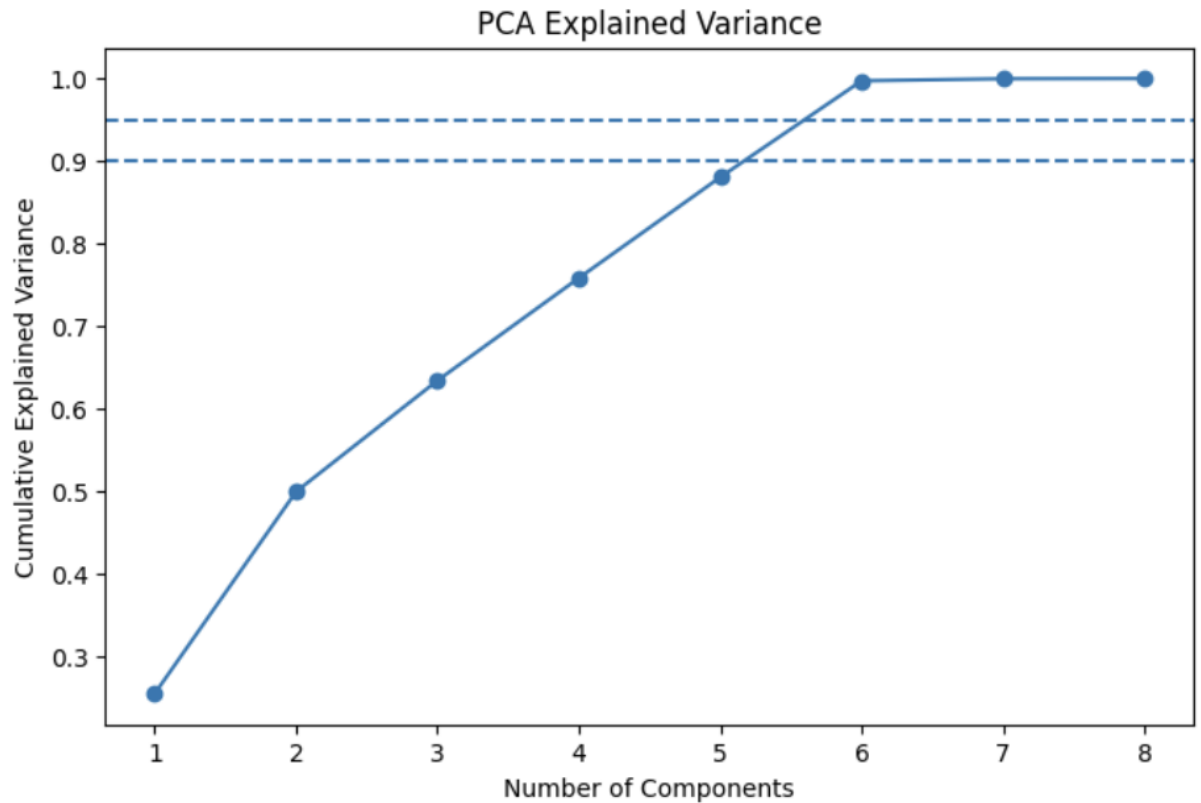
- First 5 components ~88% of variance.
- Around 6 components ~95%+ of variance.
- PCA reduced model dimensions from 25 → 14

PCA vs Original

- Original features achieve $R^2 = 0.855$ while PCA features achieve $R^2 = 0.841$

Key Takeaway

- PCA reduces complexity with minimal performance loss, but original features perform slightly



Regression Model Performance

Models Tested

- Linear Regression, Ridge, Lasso, SVR (Support Vector Regression)
- Target variable: log_charges
- Evaluation metrics: MAE, RMSE, R^2 , 5-fold CV R^2

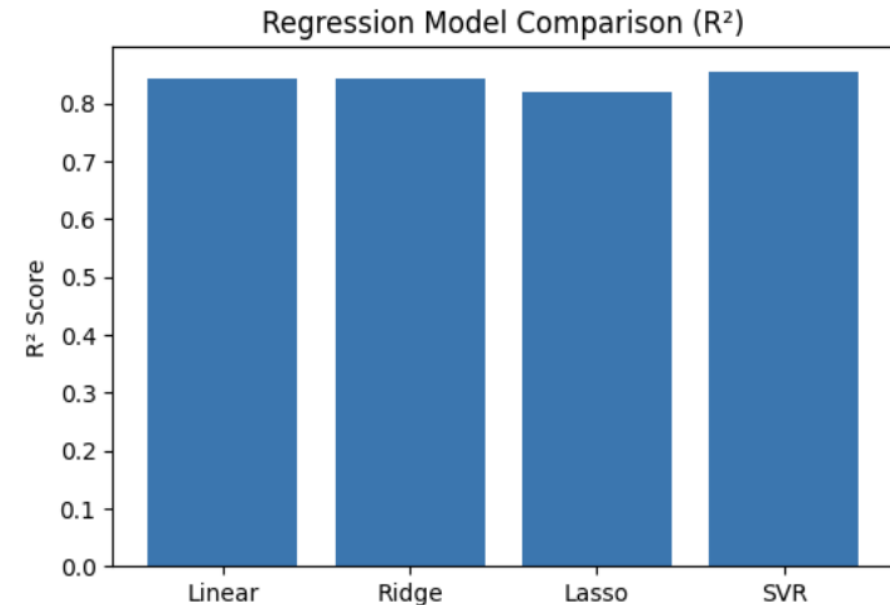
Cross-Validation Insight

- Linear/Ridge have the highest mean CV R^2 : ~ 0.770
- SVR has the best test R^2 : 0.855
- Lasso performs weakest overall

Key Takeaway

- SVR captures non-linear relationships best on the test set, while Ridge/Linear show strong stability in cross-validation

Model	MAE	RMSE	R2
Linear	0.2206	0.3553	0.8418
Ridge	0.2205	0.3551	0.8419
Lasso	0.2411	0.3785	0.8204
SVR	0.2163	0.3402	0.855



Classification Model Performance

Models Tested

- Logistic Regression
- SVM (Support Vector Machine) with RBF kernel
- MLP (Multi-Layer Perception Neural Network)

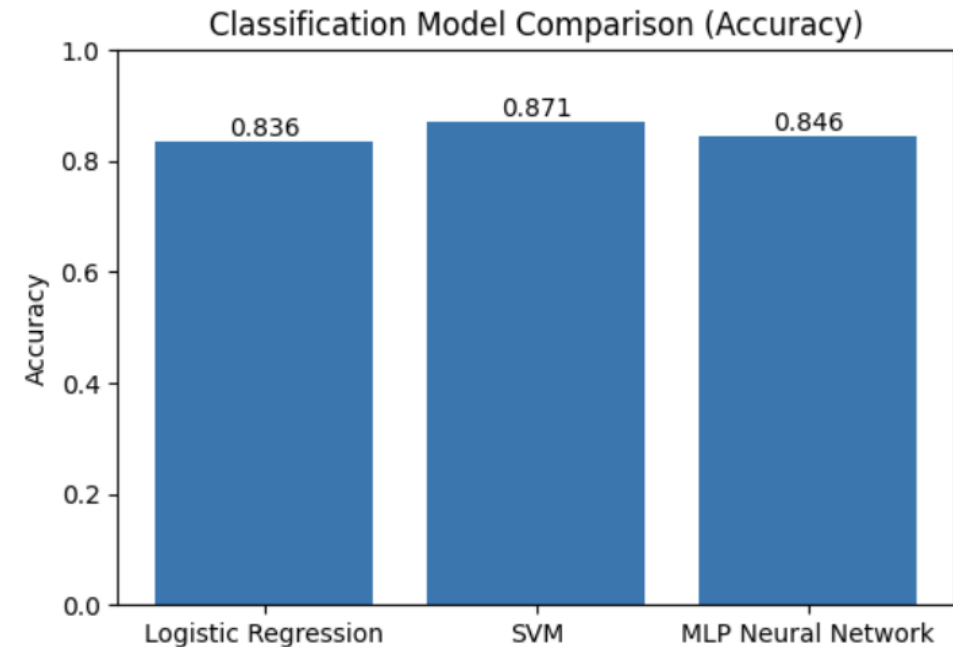
Learning Curve Insight

- MLP training accuracy increases to about 93%
- Validation accuracy stabilizes around 86 - 87%
- Small gap suggests mild overfitting, but not severe

Key Takeaway

- SVM performs best overall, while MLP shows strong learning behavior but slightly lower validation/test performance.

Model	Accuracy	Macro F1	AUC-ROC
Logistic Regression	0.8358	0.8354	0.9397
SVM	0.8706	0.8709	0.9558
MLP Neural Network	0.8458	0.8448	0.9427



Model Performance – Confusion Matrix Analysis

Compared class-wise prediction :

- Logistic Regression , SVM , MLP Neural Network

Key Observations

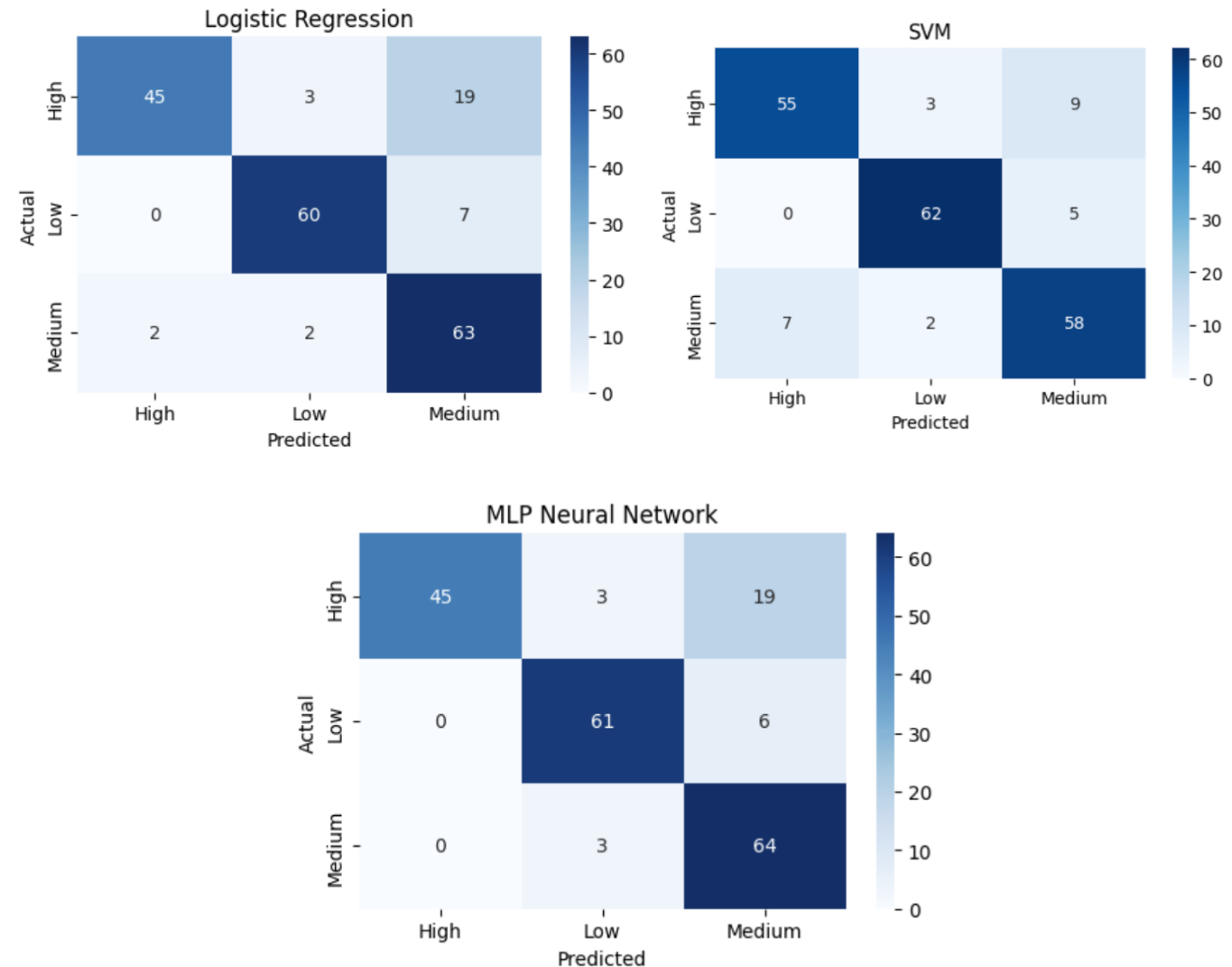
- **SVM performs best overall** : High: 55 correct , Low: 62 correct , Medium: 58 correct
- **Logistic Regression** : Strong for Low (60 correct), Misclassifies High → Medium (19 cases)
- **MLP Neural Network** : Strong for Medium (64 correct), Similar confusion: High → Medium (19 cases)

Common Error Pattern

- High-risk individuals often predicted as Medium risk
- Indicates overlap between High vs Medium features

Key Takeaway

- SVM achieves the most reliable and balanced classification, while all models struggle slightly to distinguish high-risk from medium-risk individuals.



Advanced Model Comparison: Random Forest vs XGBoost

Compare ensemble models for regression & classification

Regression Performance

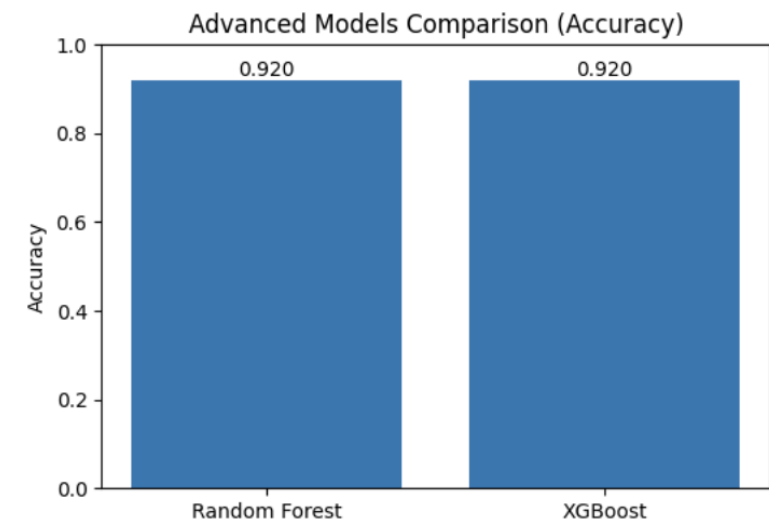
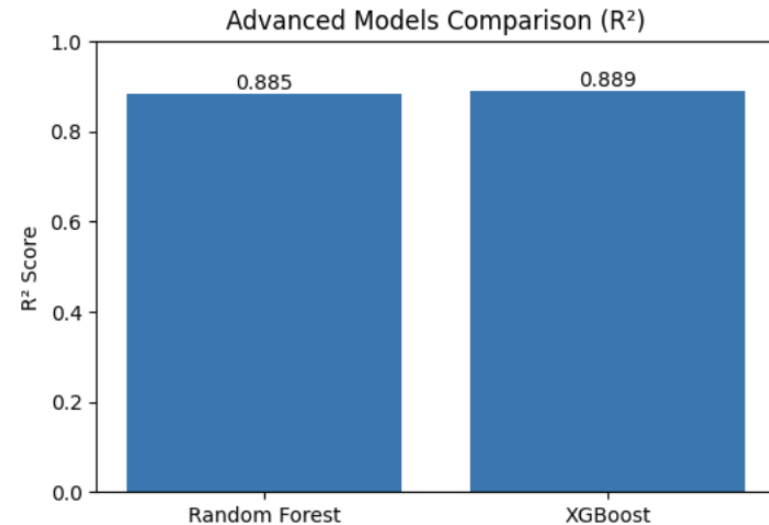
- XGBoost performs best
- R^2 : 0.889 vs 0.885
- MAE: 0.159 vs 0.165
- High accuracy (~92%)

Nearly Classification Performance

- Both models achieve very identical Macro F1 → balanced predictions across classes

Key Takeaway

- XGBoost is the best overall model for regression tasks
- For classification, both models are equally strong and stable
- Ensemble methods significantly outperform traditional models



Final Model Comparison & Selection

Objective

Identify the best-performing models for:

- Regression (insurance cost prediction)
- Classification (risk tier prediction)

Regression Performance (R^2)

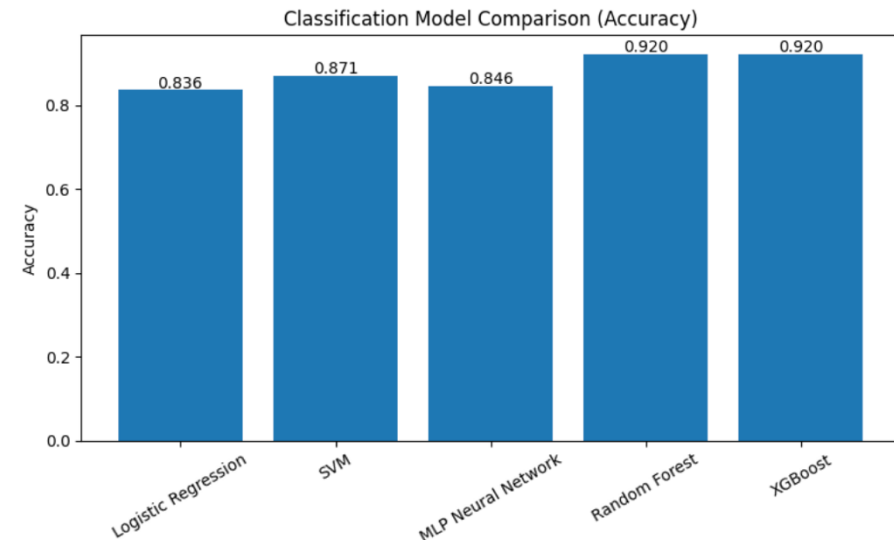
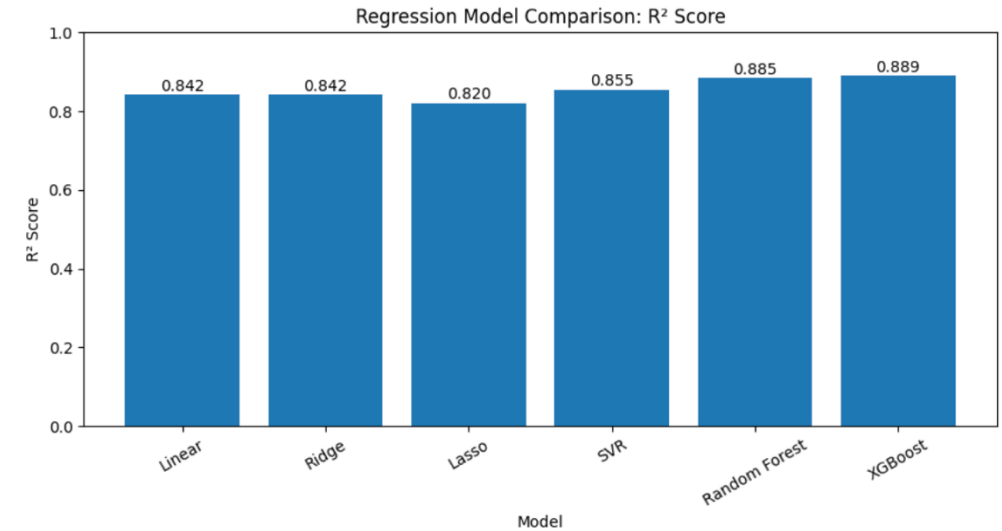
- XGBoost: 0.889 (Best), Random Forest: 0.885, SVR: 0.855, Linear / Ridge: ~0.842, Lasso: 0.820 (Lowest)

Classification Performance (Accuracy)

- Random Forest & XGBoost: ~0.920 (Best), SVM: 0.871, MLP: 0.846, Logistic Regression: 0.836

Key Takeaway

- Ensemble models outperform all others, with XGBoost leading regression and Random Forest/XGBoost dominating classification. Overall, models deliver high accuracy with strong and stable performance across tasks.



Conclusion

Key Findings

- Feature engineering significantly improved model performance
- Log transformation reduced skewness and stabilized predictions
- Ensemble models captured complex patterns better than linear models

Model Performance

- XGBoost achieved highest regression performance ($R^2 = 0.889$)
- Random Forest & XGBoost achieved highest classification accuracy (~92%)
- Traditional models performed well but were outperformed by ensemble methods

Insights

- Smoking-related and risk-based features are strong predictors
- Some overlap exists between High vs Medium risk categories
- PCA reduced dimensionality but slightly reduced accuracy